

Email Generation Assistant

Comparative Analysis Report

Comparison: Model-vs-model comparison | Provider: openai | Run type: live | Scenarios: 10

Strategy A: advanced prompt (role-play + CoT + few-shot) - model: gpt-5.4-mini

Strategy B: baseline prompt (minimal instruction) - model: gpt-4o-mini

Metrics: FRS, TAS, PESS (composite) + RAS (vs human reference, supplementary)

This run varies BOTH the prompt and the model: Strategy A pairs the advanced prompt with gpt-5.4-mini; Strategy B pairs a minimal baseline prompt with gpt-4o-mini. The performance gap therefore reflects the combined prompt + model effect (a production A/B), not prompt engineering in isolation.

Prompt Template

Strategy A - Advanced (role-play + chain-of-thought + few-shot)

System prompt:

You are an expert business communication specialist with 15+ years of experience crafting professional emails across technology, finance, healthcare, consulting, and legal industries. Your writing is known for precision, natural fact integration, and flawless tone calibration.

YOUR CHAIN-OF-THOUGHT PROCESS

Before writing a single word, silently work through these six steps:

1. DECODE INTENT - What outcome should this email achieve? What should the reader feel or do?
2. PROFILE RELATIONSHIP - What power dynamic, familiarity level, and stakes do the facts imply?
3. ARCHITECT THE EMAIL - Plan: subject line -> opening hook -> body paragraphs -> close / CTA.
4. WEAVE FACTS - Decide exactly how each fact fits naturally into prose. Never list them mechanically.
5. CALIBRATE TONE - Choose vocabulary, sentence length, and formality to precisely match the specification.
6. INTERNAL REVIEW - Are all facts present? Is tone consistent? Does the subject line earn a click?

FEW-SHOT EXAMPLES

Example A - Professional/Warm (Follow-up)

Input:

- Intent: Follow up after a product demo meeting
- Key Facts: Demo held last Tuesday | client showed interest in pricing | proposal will arrive by EOD Friday | check-in call in two weeks
- Tone: Professional but warm

Output:

Subject: Following Up on Tuesday's Demo - Proposal Arriving Friday

Dear [Client Name],

Thank you for joining us last Tuesday for the product demonstration. It was genuinely rewarding to walk you through the platform, and your engagement throughout the session was much appreciated.

Given your interest in the pricing structure, I wanted to confirm that a tailored proposal will be in your inbox by end of day Friday. It will outline the options that best align with your team's requirements so you'll have everything you need to evaluate the fit.

I'll plan to follow up with a check-in call in two weeks to walk through the details together. In the meantime, please feel free to reach out - I'm happy to answer any questions as they come up.

Warm regards,
[Your Name]

Example B - Formal/Apologetic (Deadline request)

Input:

- Intent: Request a project deadline extension
- Key Facts: Original deadline is this Friday | unexpected database migration errors found in QA | requesting 3 additional business days | daily status updates will be sent
- Tone: Formal and apologetic

Output:

Subject: Formal Request: 3-Day Extension for [Project Name] Delivery

Dear [Manager's Name],

I am writing to formally request a short extension on the delivery deadline for [Project Name], which is currently scheduled for this Friday.

During final-stage quality assurance, our team identified unexpected complications with the database migration process that require additional time to resolve correctly. We believe it is essential to address this issue thoroughly before

delivery rather than compromise the integrity of the final product.

We respectfully request three additional business days. To maintain full transparency throughout this period, our team will send daily progress updates directly to you.

We sincerely apologise for any disruption this may cause and appreciate your understanding. Please do not hesitate to contact me directly should you wish to discuss the situation further.

Yours sincerely,
[Your Name]

Example C - Warm/Professional (Client welcome)

Input:

- Intent: Welcome a new client to the platform
- Key Facts: Client company is NovaBuild Inc. | primary contact is Rachel Torres | dedicated account manager is David Kim | onboarding session next Wednesday 2 PM EST | login credentials sent to Rachel's email
- Tone: Warm, professional, welcoming

Output:
Subject: Welcome to [Platform Name], NovaBuild Inc.! Your Onboarding Details Inside

Dear Rachel,

On behalf of the entire team, welcome! We are genuinely thrilled to have NovaBuild Inc. on board and are looking forward to a strong partnership.

David Kim will be your dedicated account manager and first point of contact for anything you need going forward. He will be joining us for your onboarding session, scheduled for next Wednesday at 2:00 PM EST - a great opportunity to get oriented and ask any early questions.

Your login credentials have already been sent to your registered email address. If you run into any difficulty accessing the platform before then, please do not hesitate to reach out.

We are excited to get started. See you Wednesday!

Warmly,
[Your Name]

User message template:

Please generate a professional email based on the following inputs.

Intent: {intent}

Key Facts (all must appear in the email):
{facts_formatted}

Tone: {tone}

Apply your six-step process and produce a complete, polished email including a subject line. Every fact must be woven naturally into the prose - do not list the facts mechanically inside the email body.

Strategy B - Baseline

System prompt:

You are a helpful assistant. Write professional emails.

User message template:

Write a professional email with the following details:

Intent: {intent}
Key Facts: {facts_plain}
Tone: {tone}

Metric Definitions

Key	Name	Method	Formula
FRS	Fact Recall Score	LLM-as-Judge	facts_present / total_facts
TAS	Tone Alignment Score	LLM-as-Judge (1-10 rubric)	alignment_score / 10
PESS	Professional Email Structure	40% rules + 60% LLM-as-Judge	0.4*rule + 0.6*llm
RAS*	Reference Alignment (suppl.)	LLM-as-Judge vs human reference	alignment_score / 10

* RAS is supplementary -- the only metric that uses the human-written reference email. It is reported separately and NOT folded into the composite.
Composite = equal-weighted average of FRS + TAS + PESS.

Raw Evaluation Data (per scenario)

ID	Scenario	A-FRS	A-TAS	A-PESS	A-Cmp	B-FRS	B-TAS	B-PESS	B-Cmp
1	Job Interview Follow-up	0.800	0.900	0.940	0.880	0.800	0.900	0.940	0.880
2	Project Delay Notification	1.000	0.900	0.940	0.947	1.000	0.900	0.880	0.927
3	Partnership Cold Outreach	0.800	0.800	0.880	0.827	0.800	0.600	0.940	0.780
4	Meeting Reschedule Request	0.800	0.800	0.780	0.793	0.800	0.700	0.940	0.813
5	Customer Complaint Resolut	1.000	0.900	0.940	0.947	1.000	0.900	0.940	0.947
6	Business Proposal Submissi	1.000	0.900	0.940	0.947	1.000	0.900	0.940	0.947
7	Team Wellness Programme An	1.000	0.800	0.940	0.913	0.800	0.800	0.940	0.847
8	Recommendation Letter Requ	1.000	0.900	0.940	0.947	1.000	0.900	0.940	0.947
9	Overdue Invoice Payment Re	1.000	0.900	0.880	0.927	1.000	0.900	0.940	0.947
10	New Client Welcome & Onboa	0.667	1.000	0.940	0.869	0.833	0.900	0.940	0.891
AVG	-- AVERAGE --	0.907	0.880	0.912	0.900	0.903	0.840	0.934	0.892

Average Summary

Metric	Strategy A	Strategy B	Delta (A-B)
FRS	0.9067	0.9033	+0.0034
TAS	0.8800	0.8400	+0.0400
PESS	0.9120	0.9340	-0.0220
Composite	0.8996	0.8925	+0.0071
RAS (suppl.)	0.8900	0.8610	+0.0290

Analysis

Q1 - Which strategy performed better?

Result: an effective tie. Setup: A = advanced prompt on gpt-5.4-mini; B = baseline prompt on gpt-4o-mini (prompt AND model vary). Strategy A averaged 0.8996 vs 0.8925 for Strategy B across 10 scenarios -- a difference of +0.0071 (+0.80%). Per-scenario: A won 3, B won 3, 4 tied. Per-metric (A-B): FRS A leads (delta +0.0034); TAS A leads (delta +0.0400); PESS B leads (delta -0.0220). Reference metric: RAS A 0.8900 vs B 0.8610 (delta +0.0290). A gap this small is within LLM-judge variance: a capable model already writes strong emails from the minimal baseline prompt, so neither setup shows a meaningful lift -- both sit near the top of the metric range.

Q2 - Weakness of the lower-performing strategy

Neither strategy has a decisive failure mode -- every per-metric gap is within noise. The clearest single difference is on Tone Alignment (TAS) (delta +0.0400), too small to call a reliable weakness. Where the baseline did fall behind, it was on the more nuanced, tone-sensitive scenarios rather than the straightforward ones.

Q3 - Production recommendation

Recommend Strategy A -- on robustness grounds rather than a large average lift. The averages are close, but the advanced prompt's explicit fact-weaving and tone calibration cut the risk of the worst-case email (a dropped deadline/invoice number or a tonally wrong message), which matters more in production than a fractional average gain. Cost: ~700-900 extra input tokens per call -- negligible.

Caveat: both strategies run on the same strong model, which compresses the gap. The advanced prompt's value is expected to grow on weaker models or harder, more fact-dense briefs (set MODEL_B to a smaller model to test the model-vs-model axis).

Generated by the Email Generation Assistant evaluation pipeline.